

Welcome to The Physics of Climate Change!

► Syllabus

- Course organization and assignments
- Grading and Evaluation
- Overview and Calendar of Topics

Physics of Climate Change — Winter 2026 — Tentative Course Calendar

Week	Lecture Topic (Tue)	Laboratory (Wed)	Discussion (Thu)
Jan 13–15	Temperatures and CO ₂ over Time	Measurement and Uncertainty	Timescales of Earth and Temp
Jan 20–22	Contemporary Climate Change	Global Average Temperature	Contemporary Climate Change
Jan 27–29	Energy and Equilibrium	Energy and Equilibrium (leaky buckets)	Energy Flow and Equilibrium
Feb 3–5	EM Radiation (light) and Blackbody	EM Rad and Blackbody Rad	BBody Rad & Earth Temp
Feb 10–12	Atmosphere and Greenhouse Eff I	Heating by radiation	Earth temp: w/Atmosphere
Feb 17–19	Atmosphere and Greenhouse Eff II	Light Absorption and Transmission	Atmosph Absorption Spectra
Feb 24–26	Atmosph Absorption Spectra	Evaporation by boiling	IR Absorp Spectra
Mar 3–5	Water and Phase Transitions	Evaporation and Clouds	Evaporation & RC Equil
Mar 10–12	Spring Break (no class)		
Mar 17–19	Paleo Perturbations of Temperature	Milankovitch Cycles	Milankovitch
Mar 24–26	Climate Sensitivity	IR absorption by GH gases	Climate Sensitivity
Mar 31– 2	Timescales of Climate Change	Equilibration Times	Rates of Change, Timescales
Apr 7–9	Modeling Climate Change	Computational Climate Models	
Apr 14–16	Oceans and Ice	Ice and Thermal Exp of H ₂ O	Oceans and Ice
Apr 21–23	work on project	IPCC Reports & Climate News	work on project
FINAL EXAM TIMESLOT: Tuesday, April 28, 2025, 4–6 pm (Poster Presentations)			

Climate Change is (mostly) a story of *Temperature*

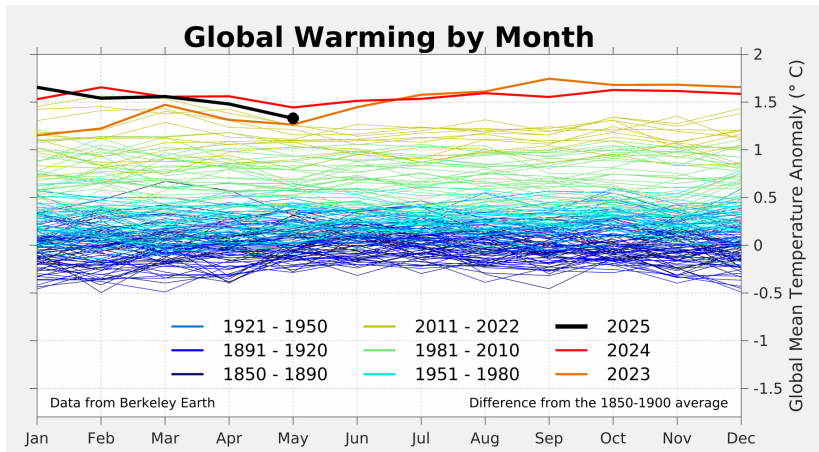


Image and data From [Berkeley Earth May 2025 update](#)
[NASA "Climate Spiral" Animation \(1880–2021\)](#)

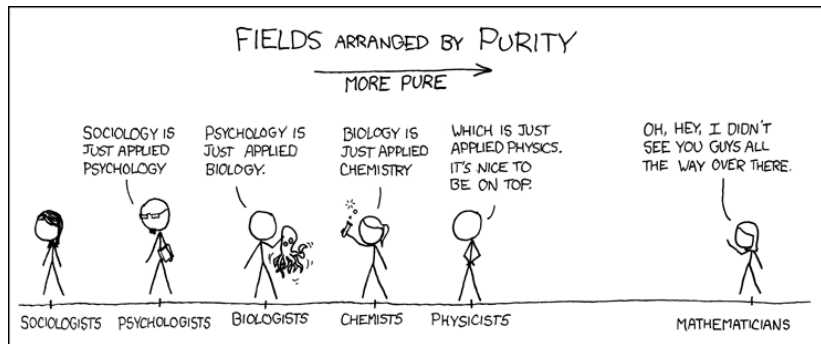
But what *is* Temperature? It is a physical quantity, so ...

What is Physics???

Physics is the precise characterization of natural phenomena, enabled by measurement and described by theoretical (mathematical) models.

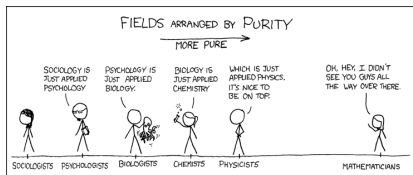
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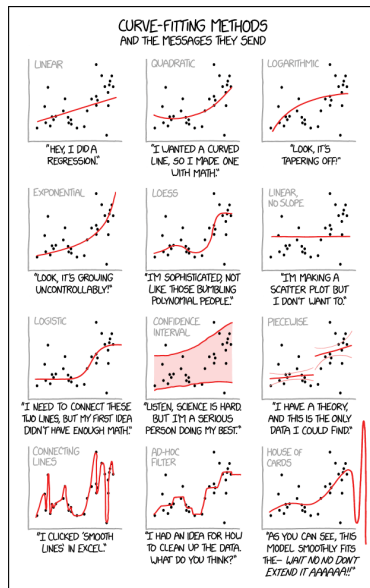
O Hai!,

PHILOSOPHERS

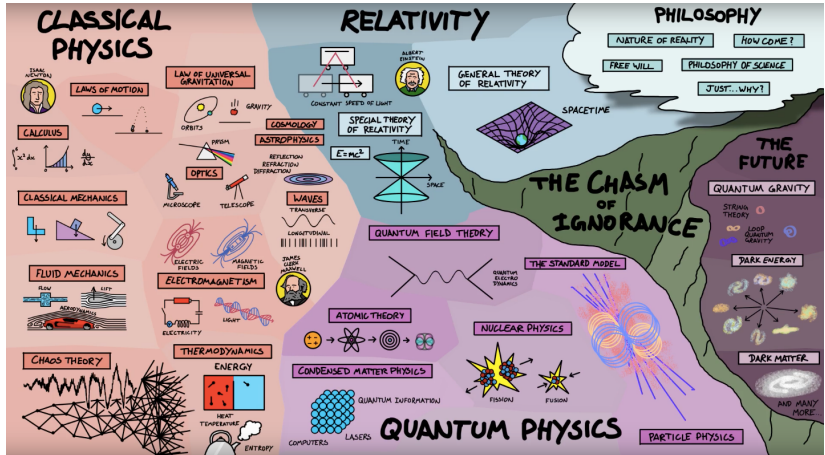


Data, Theoretical Models, and Uncertainty

- ▶ “dots” (*measurement values*) are the job of the experimental physicist
- ▶ developing a “line” (*theoretical/mathematical model*) is the job of a theoretical physicist
- ▶ Each dot should actually have “bars” to indicate the *uncertainty* of each measurement
- ▶ A “good” theory is the simplest line that goes through the dots, taking into account the uncertainties

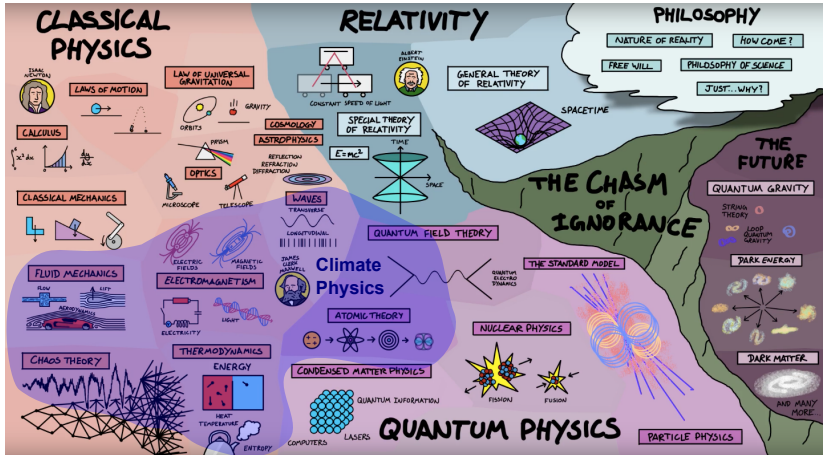


What is Physics: Sub-disciplines



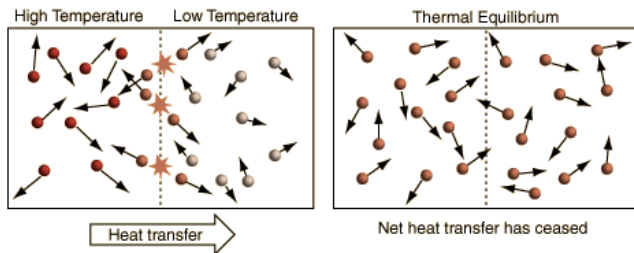
"The Map of Physics" by Dominic Walliman — 2016 youtube video

What is Physics: Sub-disciplines



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Thermodynamics: Energy, Temperature, and Equilibrium

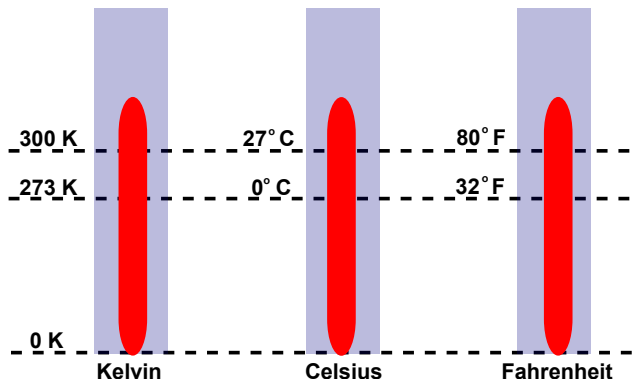


From *Hyperphysics — A More General View of Temperature*

- ▶ The *internal energy* of an object is the random “thermal” motion of its constituent particles.
- ▶ *Temperature* is a measure of (proportional to) that energy
- ▶ *Thermal contact* between two objects at different temperatures will eventually result in a common temperature (*thermal equilibrium*)

Temperature Scales

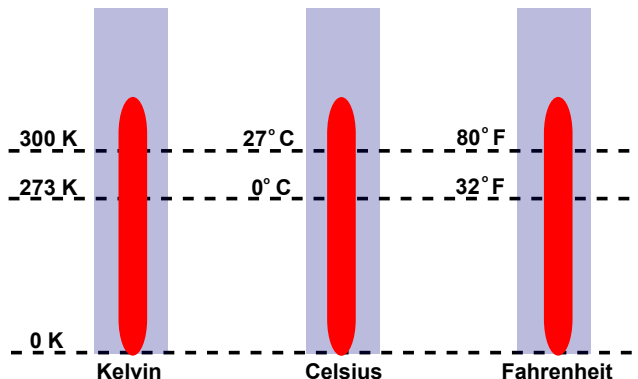
There are three common units of temperature.



- ▶ The first (Kelvin) is the physicist's standard unit
- ▶ The last (Fahrenheit) is just weird, but comfortable
- ▶ We will primarily use the second (Celsius)

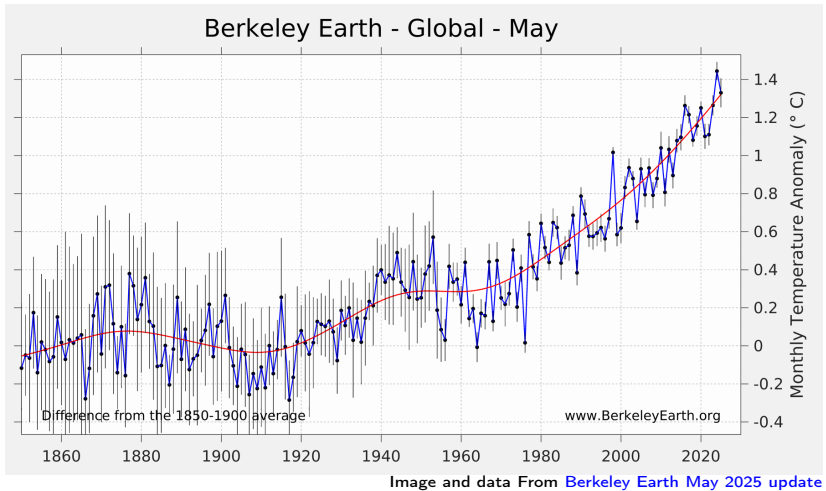
Thermometers

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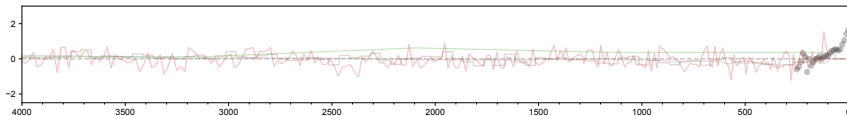
Climate Change is (pretty) *Recent*



- Note: *uncertainty* on each point and temperature *anomaly*
- Temperature increase is outside uncertainty, but perhaps just *natural variation*?

Paleo-record of temperature (proxies)

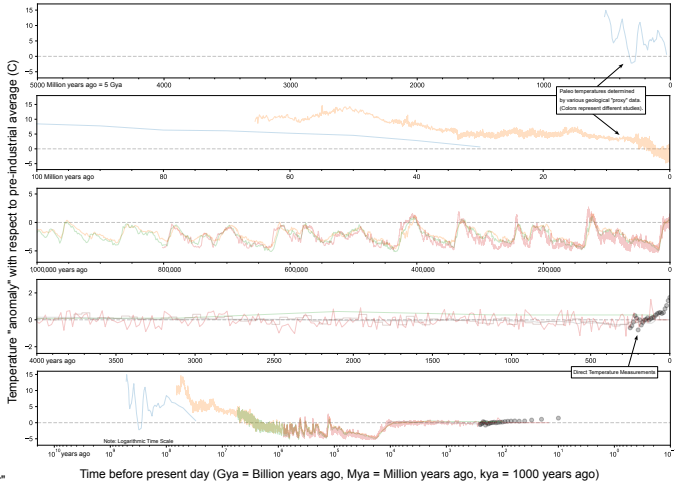
Looking at 4000 years of (proxy) temperature measurements. . .



This week in Discussion...



Segment of the
Universe's "Calendar"
(13.8 Billion years old)



(Proxy) Temperature record over the history of the Earth